

Executive Summary Report

Delivery Time Estimation Using Linear Regression - Porter

July 11, 2025

1. Objective

This project analyzes delivery time data using Linear Regression. The goal is to identify key variables that influence delivery time such as subtotal, distance, dasher availability, and order patterns, and to create a predictive model for improving delivery logistics and efficiency.

2. Methodology

1. Data cleaning and preprocessing, including datetime parsing and outlier treatment.
2. Feature engineering (hour, day, weekend).
3. Exploratory data analysis using distribution and box plots.
4. Feature selection via correlation heatmap and Recursive Feature Elimination (RFE).
5. Model building using linear regression.
6. Evaluation using residual analysis and impact of scaling.

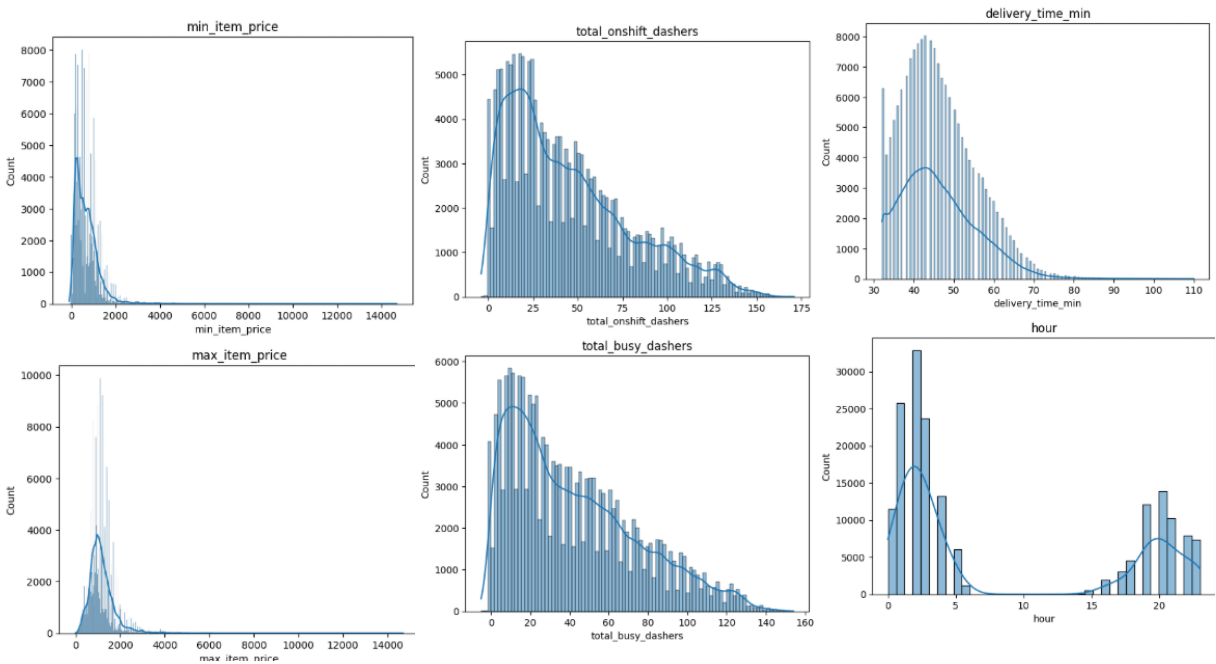
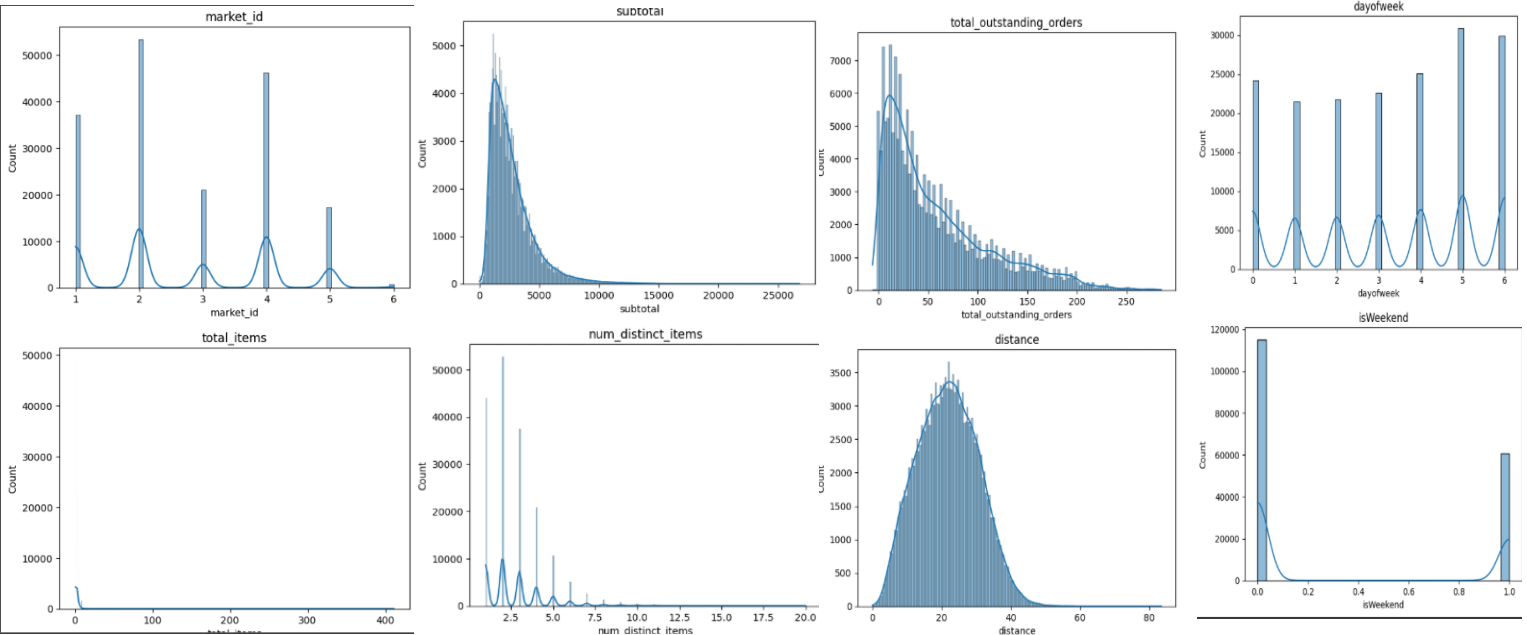
2. Key Insights

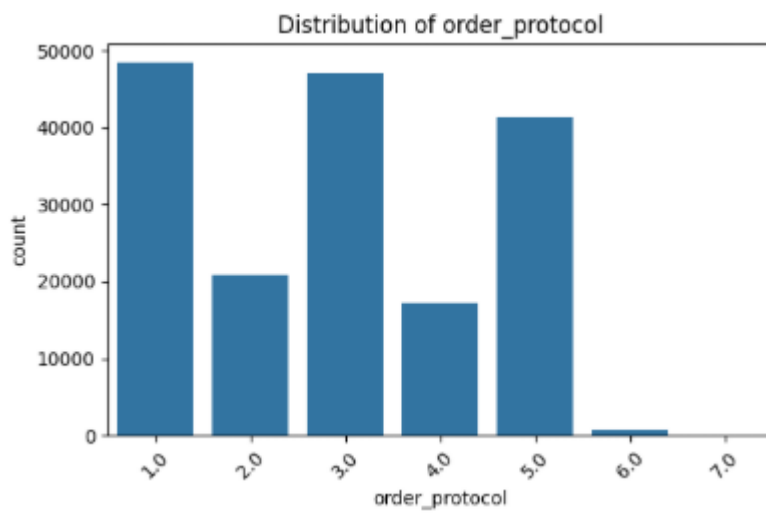
- Delivery time is affected by item count, subtotal, and distance; most features are right-skewed, especially delivery time.
- Order protocols 1, 3, and 5 are most common.
- Delivery time is right-skewed; most deliveries take 35–50 mins, with few delays due to demand or traffic
- Delivery time is most influenced by distance, subtotal, and dasher availability.
- Late-night orders take longer, likely due to lower dasher availability.
- Strong positive correlation exists between delivery time and distance.
- Delivery times are typically between 35–50 minutes, with some longer outliers.
- Residual plots indicate a mostly linear and unbiased prediction performance.
- Feature scaling helps models treat variables equally and improves accuracy.
- Delivery time rises with distance; each unit adds ~0.48 minutes.

3. Supporting Visualizations

The following plots support the data insights and modeling findings:

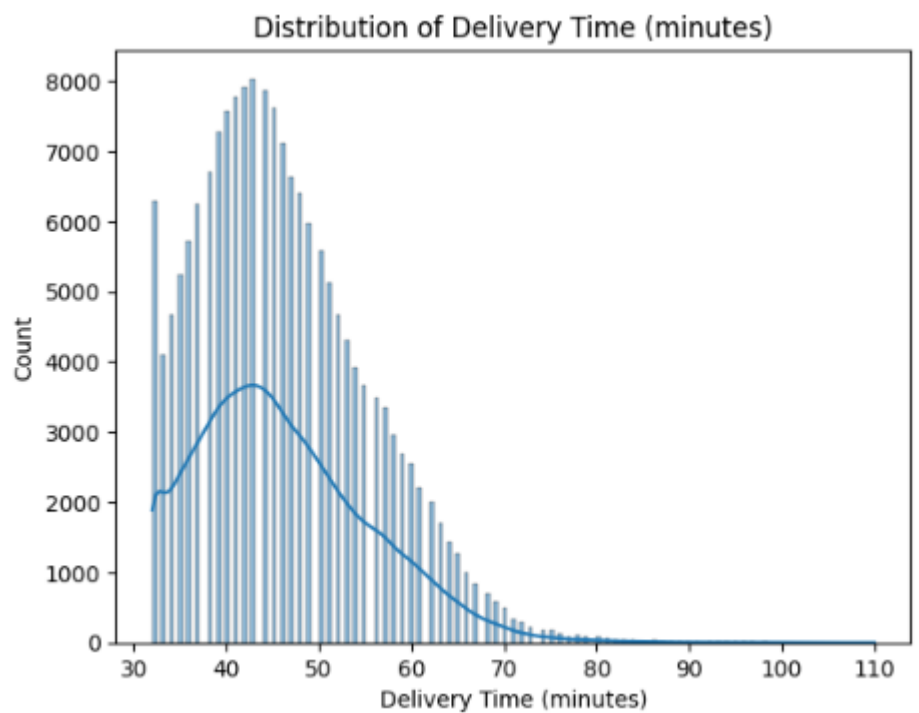
- **Delivery time is impacted by item count, subtotal, and distance; most features are right-skewed, especially delivery time.**



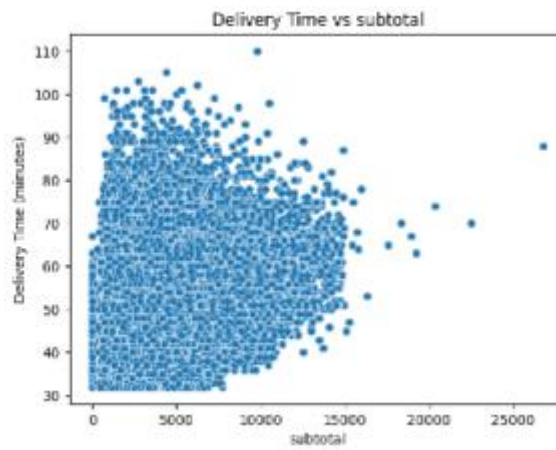


- Order protocols 1, 3, and 5 dominate; protocol 6 is rare and may impact delivery time differently.

- Delivery time is right-skewed; most deliveries take 35–50 mins, with few delays due to demand or traffic.

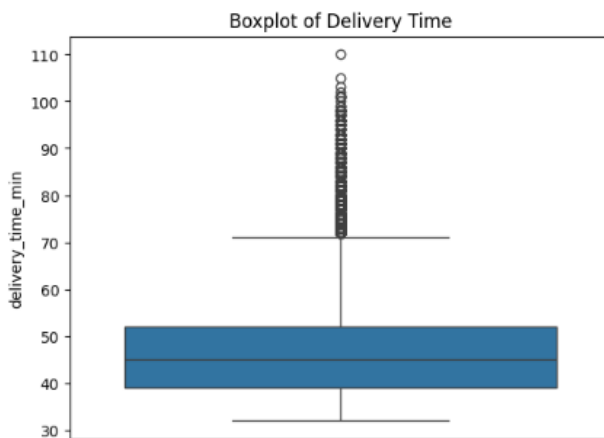


- Delivery time is most influenced by distance, subtotal, and dasher availability.

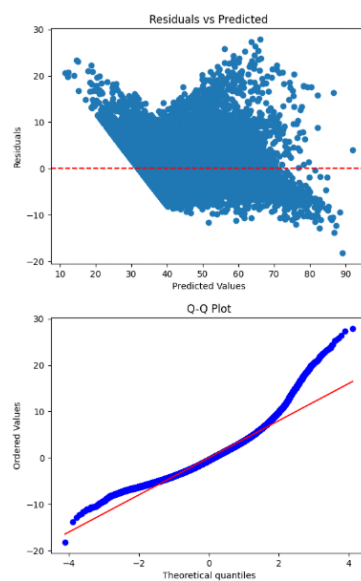
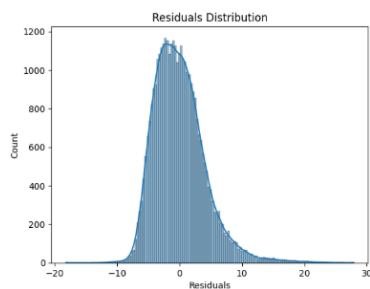


- Late-night orders take longer, likely due to lower dasher availability.

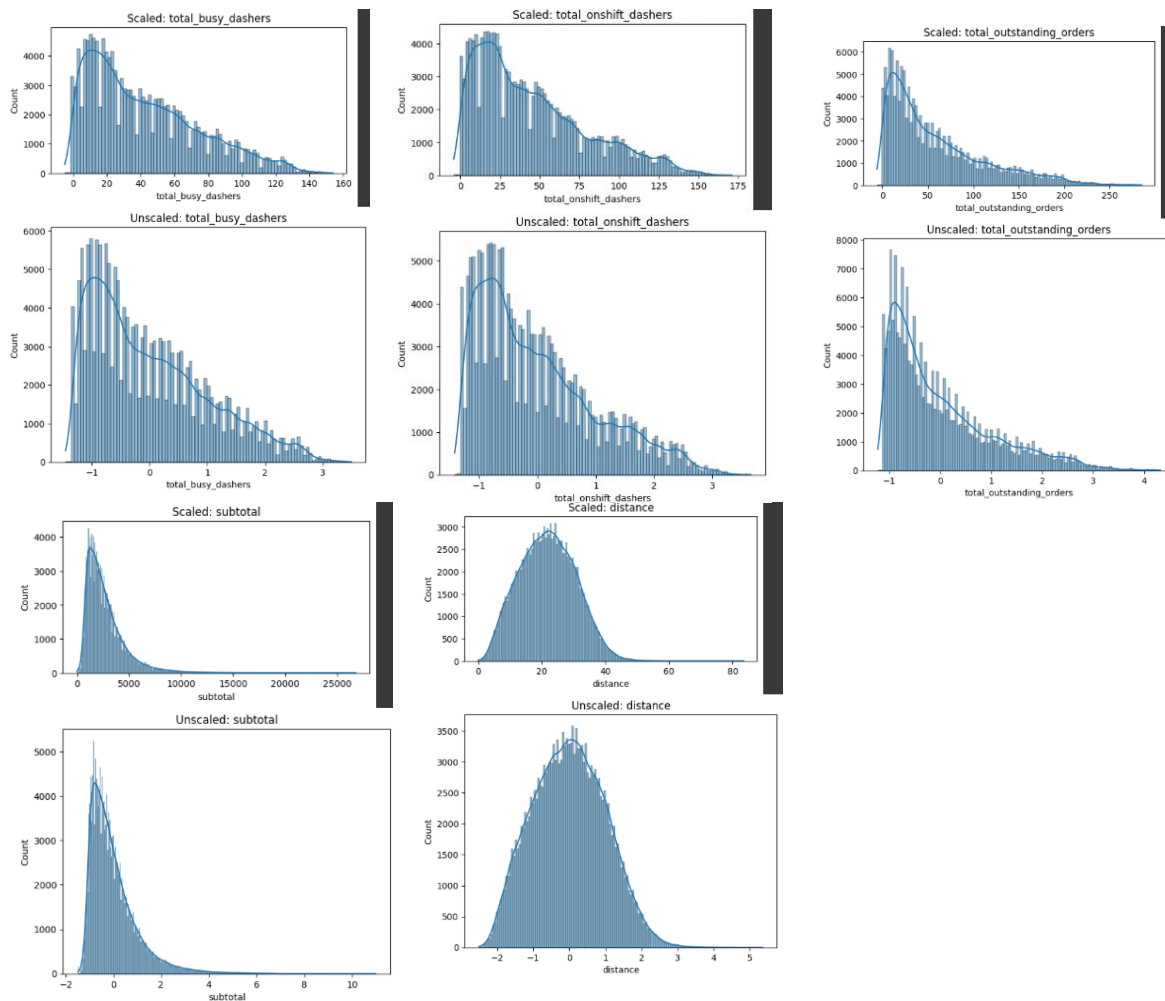
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- Correlation Heatmap
- | | market_id | total_items | subtotal | num_distinct_items | min_item_price | max_item_price | total_onshift_dashers | total_busy_dashers | total_outstanding_orders | distance | delivery_time_min | hour | dayofweek | isWeekend |
|--------------------------|-----------|-------------|----------|--------------------|----------------|----------------|-----------------------|--------------------|--------------------------|----------|-------------------|-------|-----------|-----------|
| market_id | 1.00 | 0.00 | -0.00 | 0.02 | -0.01 | -0.01 | 0.07 | 0.07 | 0.07 | 0.02 | -0.07 | -0.01 | -0.00 | -0.00 |
| total_items | 0.00 | 1.00 | 0.55 | 0.76 | -0.39 | -0.05 | 0.03 | 0.03 | 0.03 | 0.01 | 0.22 | -0.07 | 0.02 | 0.03 |
| subtotal | -0.00 | 0.55 | 1.00 | 0.68 | 0.04 | 0.51 | 0.13 | 0.13 | 0.13 | 0.04 | 0.41 | -0.19 | 0.03 | 0.05 |
| num_distinct_items | -0.02 | 0.76 | 0.68 | 1.00 | -0.45 | 0.05 | 0.07 | 0.06 | 0.07 | 0.02 | 0.31 | -0.12 | 0.03 | 0.04 |
| min_item_price | -0.01 | -0.39 | 0.04 | -0.45 | 1.00 | 0.54 | 0.04 | 0.04 | 0.04 | 0.00 | 0.02 | -0.05 | -0.00 | 0.00 |
| max_item_price | -0.01 | -0.05 | 0.51 | 0.05 | 0.54 | 1.00 | 0.13 | 0.13 | 0.13 | 0.03 | 0.26 | -0.19 | 0.03 | 0.05 |
| total_onshift_dashers | -0.07 | 0.03 | 0.13 | 0.07 | 0.04 | 0.13 | 1.00 | 0.94 | 0.94 | 0.05 | 0.17 | -0.38 | 0.10 | 0.09 |
| total_busy_dashers | -0.07 | 0.03 | 0.13 | 0.06 | 0.04 | 0.13 | 0.94 | 1.00 | 0.93 | 0.04 | 0.21 | -0.35 | 0.09 | 0.11 |
| total_outstanding_orders | -0.07 | 0.03 | 0.13 | 0.07 | 0.04 | 0.13 | 0.94 | 0.93 | 1.00 | 0.04 | 0.38 | -0.36 | 0.09 | 0.12 |
| distance | -0.02 | 0.01 | 0.04 | 0.02 | 0.00 | 0.03 | 0.05 | 0.04 | 0.04 | 1.00 | 0.46 | -0.03 | 0.01 | 0.01 |
| delivery_time_min | -0.07 | 0.22 | 0.41 | 0.31 | 0.02 | 0.26 | 0.17 | 0.21 | 0.38 | 0.46 | 1.00 | -0.35 | 0.05 | 0.14 |
| hour | -0.01 | -0.07 | -0.19 | -0.12 | -0.05 | -0.19 | -0.38 | -0.35 | -0.36 | -0.03 | -0.35 | 1.00 | 0.01 | 0.00 |
| dayofweek | -0.00 | 0.02 | 0.03 | 0.03 | -0.00 | 0.03 | 0.10 | 0.09 | 0.09 | 0.01 | 0.05 | 0.01 | 1.00 | 0.81 |
| isWeekend | -0.00 | 0.03 | 0.05 | 0.04 | 0.00 | 0.05 | 0.09 | 0.11 | 0.12 | 0.01 | 0.14 | 0.00 | 0.81 | 1.00 |



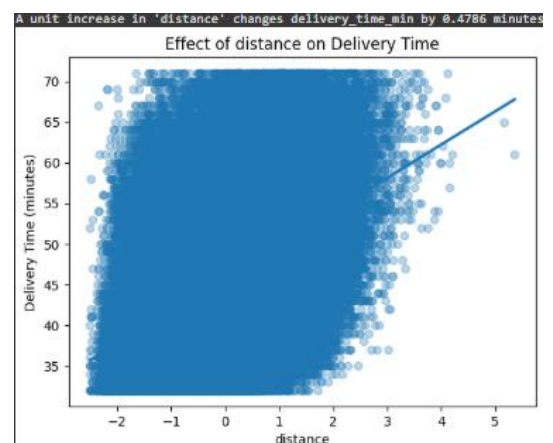
- **Residual plots indicate a mostly linear and unbiased prediction performance.**



- Feature scaling helps models treat variables equally and improves accuracy.



- Delivery time rises with distance; each unit increase adds ~0.48 minutes, confirming distance as a key factor.



5. Outcomes

The final linear regression model identifies distance, subtotal, and dasher variables as key predictors. Feature scaling improved performance, and residual analysis confirmed model reliability. These results can guide logistics planning and customer experience improvements.